

Dev Class Documentation

Source File: Dev.h
Namespace: cap
Class Header: class Dev : public Object

Overview

The *Dev* abstract class simulates a hierarchical circuit assembly with the properties:

- Each unit is a Boolean function data structure that can accept at most 10 unit inputs.
- Each unit has a range of inputs it can accept.
- Each unit can be an input of exactly one other unit.
- A unit can be a repeated input of another unit.
- A unit cannot be an input of another unit unless it is valid, which means it can successfully generate an output.
- A unit cannot be modified once it becomes an input.

Constructors

- Dev() (default constructor)
 - **Purpose:** Constructs its identification.
- Dev(const Dev& obj) (copy constructor) [deleted]

Destructor

- ~Dev() [virtual]
 - **Purpose:** Eliminates the its identification.

Assignment Operator

- operator=(const Dev& rhs) [deleted]

Member Functions

- bind(Dev& obj) [virtual]
 - **Purpose:** Makes *obj* its parent function and returns true only if it and *obj* are unbounded and it is valid; otherwise, it returns false.
 - **Parameter(s):**
 - *obj*: *Dev* reference object.
 - **Return:** Boolean.
- unbind() [virtual]
 - **Purpose:** Removes its parent function.
- reset() [virtual]
 - **Purpose:** Resets it to its default setting.
- broken() const [virtual]
 - **Purpose:** Returns true if it can be used as an input and be evaluated; otherwise, it returns false.
 - **Return:** Boolean.
- bounded() const [virtual]
 - **Purpose:** Returns true if it has a parent function; otherwise, it returns false.
 - **Return:** Boolean.
- key() const
 - **Purpose:** Returns its identification.
 - **Return:** Unsigned integer.
- valid() const [pure virtual]
 - **Purpose:** Intends to check if the function can be evaluated.
 - **Return:** Boolean.
- output() const [pure virtual]
 - **Purpose:** Intends to evaluate the function.
 - **Return:** Boolean.